FABRIC 101

Your Park Map to Microsoft Fabric

No tickets. No parking fees. No prior Fabric experience required.



Day of Data Orlando · October 17, 2026

Jason Romans · The Data Shepherd · datashepherd.dev

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Orlando, FL

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HI, I'M JASON

THE DATA SHEPHERD



Microsoft® Most Valuable Professional

Honored to be recognized by Microsoft.

I HERD THE DATA

- User Group Organizer
- Nashville, Tennessee
- Fabric + Power BI
- Teaching + Speaking







MY FAVORITE MODEL

TARDIS

TOOLS & ALLIES

Power BI | Microsoft Fabric | DuckDB | Python | SQL | VS Code

WITH THANKS

Standing on the Shoulders of Giants

Grateful for the people who share their knowledge, encourage us, answer the questions we were embarrassed to ask, and hand us the map they wish they'd had.

Nobody figures out a platform this size alone.



Take this with you

SLIDES AND DEMOS

The Data Shepherd
slides.datashepherd.dev

Slides and demos at slides.datashepherd.dev.

Session Evaluation

Fabric 101: Your Park Map to Microsoft Fabric

YOUR PARK MAP

five stops, one afternoon

1 **Main Street**
getting your bearings

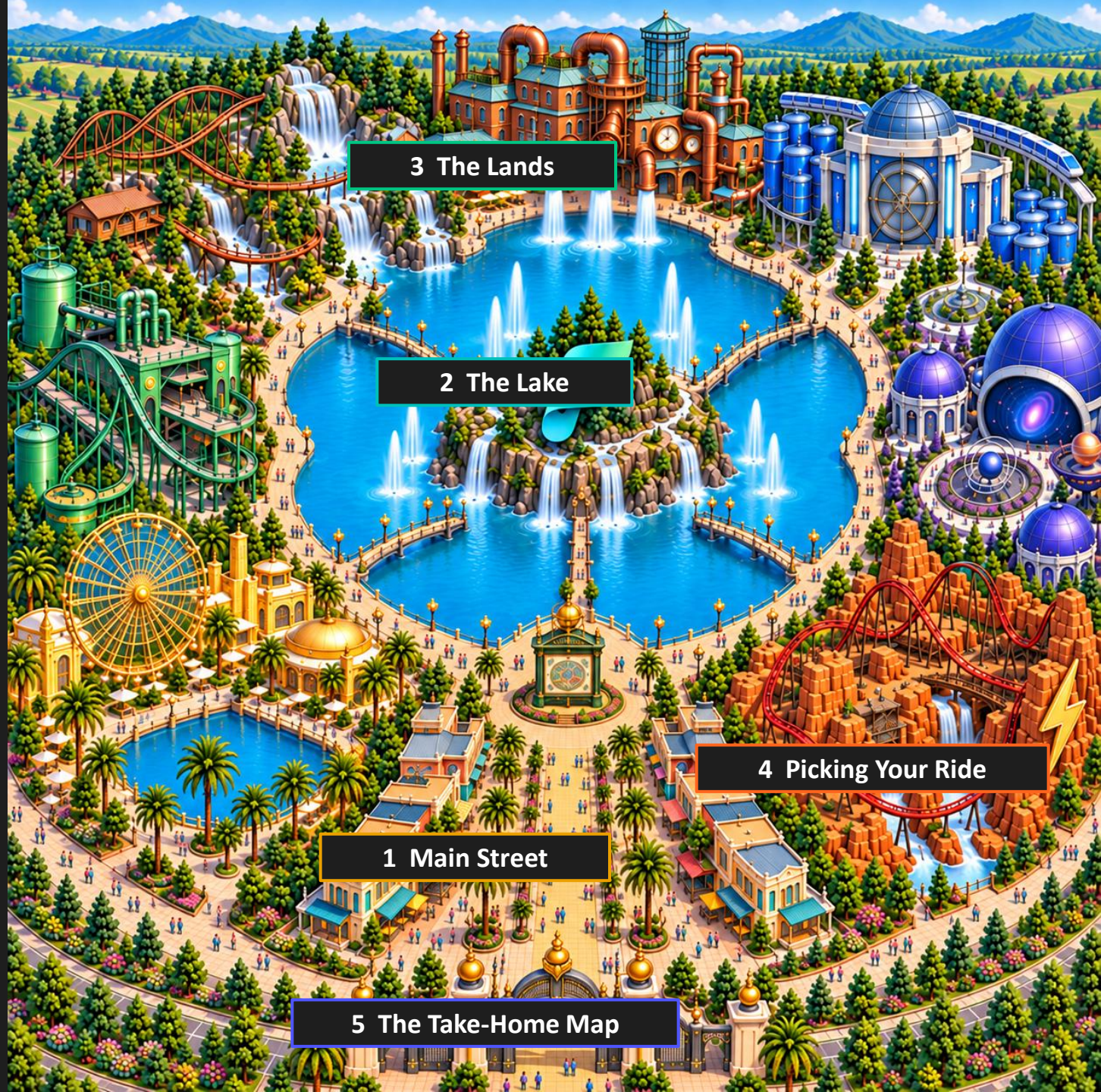
2 **The Lake**
OneLake at the centre

3 **The Lands**
all seven, one walk

4 **Picking Your Ride**
the decisions

5 **The Take-Home Map**
where to go next

One map. We move the pin, not the metaphor.



How the park works

Fabric has a lot of new words. Each one has a place on the map.

Tenant	The resort: everything your organization owns
Capacity	The power that keeps the lands open
Workload	A land: seven of them, Power BI to Databases
Items	The rides in each land: lakehouses, reports
Workspace	Backstage, where your crew builds its rides
OneLake	The Lake at the center of the park
License	Your ticket: Free gets you in. Pro lets you share.
Throttling	The lines get too long

You need a ticket, and the land has to be open.

Your ticket

Free gets you through the gate. Sharing in Power BI land is what takes Pro.

Report viewer

Watches the shows

Needs Pro or PPU, unless the park is F64 or bigger.

Power BI developer

Builds the shows

Pro to publish and share, on any park.

PPU for premium features. Everyone using PPU content needs PPU too.

Data engineer

Builds rides in the other lands

Free is enough.

Needs backstage access on Fabric capacity.

Pro to share. Your audience needs Pro too, unless the park is F64 or bigger.

Opening the lands

Your ticket gets you through the gate. Capacity decides which lands are open.

Shared capacity

Already there, free

Comes with every tenant.

Only the Power BI land is open.

Pro and PPU workspaces live here.

A trial

60 days, F4 or F64

All workloads, with feature limits.

Change size when eligible.

Test at the size you plan to buy.

Trial sizing walkthrough

Fabric capacity

An F SKU

Ask your admin if one exists.

F2 and up. Every land open.

COMING ATTRACTION · PREVIEW

F0: pay per ride, nothing to provision.
Announced at FabCon Europe.

Power BI land was built first. Capacity lights up the rest.

YOU ARE HERE

1 · MAIN STREET

Getting your bearings

What Fabric is, and why day one feels like too much.



What Fabric provides

Integrated tools for moving, shaping, analyzing and reporting on data.

The platform

Fabric brings data tools together in one service.

Power BI provides semantic models and reports.

Workspaces organize the items your team builds.

The shared lake

OneLake provides shared storage for analytical data.

Multiple Fabric tools can use the same analytical tables.

OneLake is the shared data layer. Each tool still has its own job.

Why day one feels like too much

The item list is long

Lakehouse, Warehouse, Notebook, Pipeline, Semantic Model, Eventhouse, KQL DB, Dataflow, Reflex…

The names overlap

Lakehouse and Warehouse both hold tables. Dataflow and Pipeline both move data. Nothing tells you which to pick.

The docs assume a path

Most tutorials start after the decision you are actually stuck on: which door do I walk through first?

So: start at the center

You do not need all of it. You need the hub, and one attraction that solves a problem you already have.

The Fabric portal

app.powerbi.com

A familiar entry point for Power BI reports and apps.

app.fabric.microsoft.com

Our entry point for exploring Fabric workspaces and workload items.

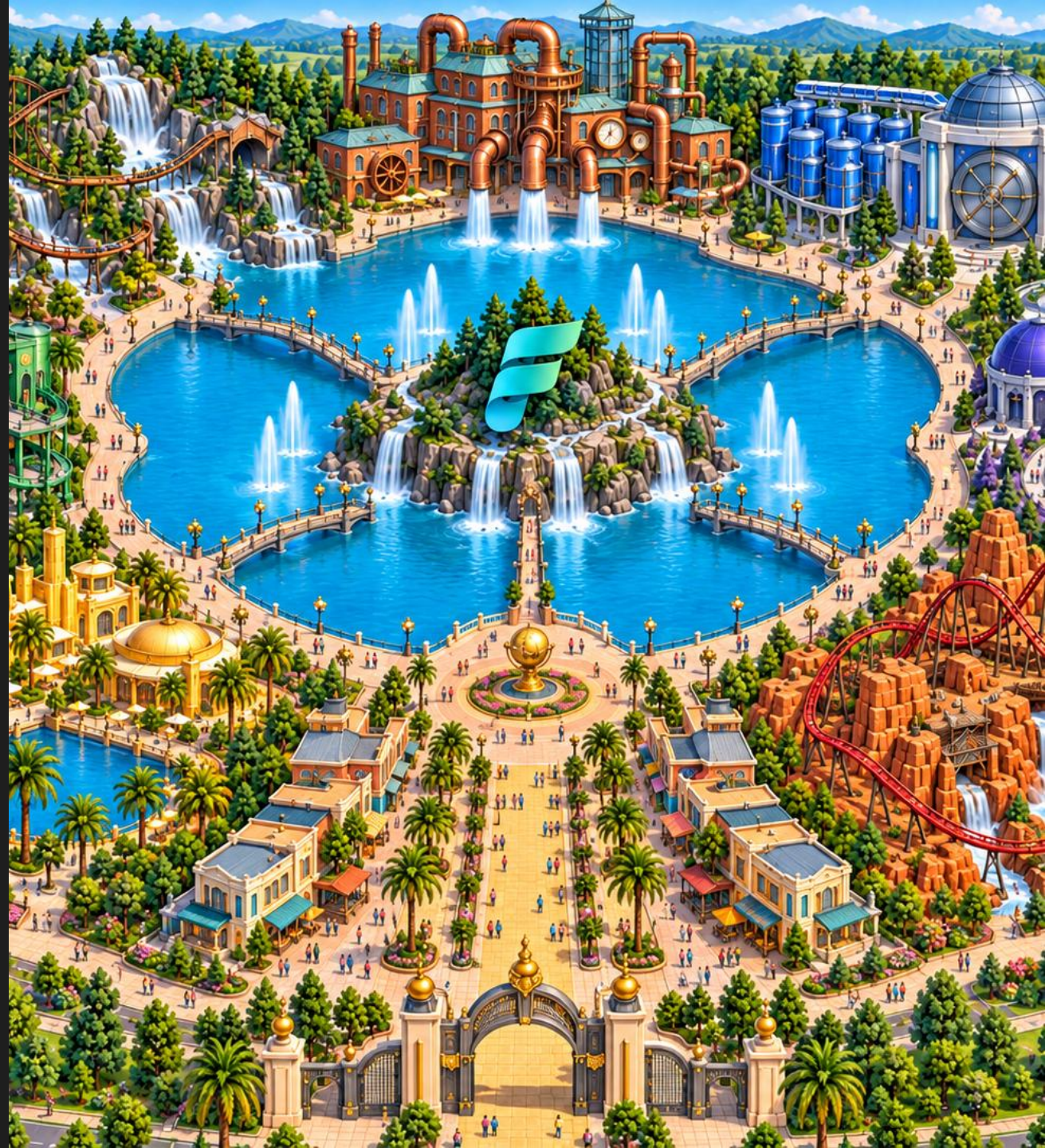
Experience picker

Use the picker in the lower-left navigation to switch between Power BI and Fabric views.

LIVE DEMO 1

Ride Along

Orientation

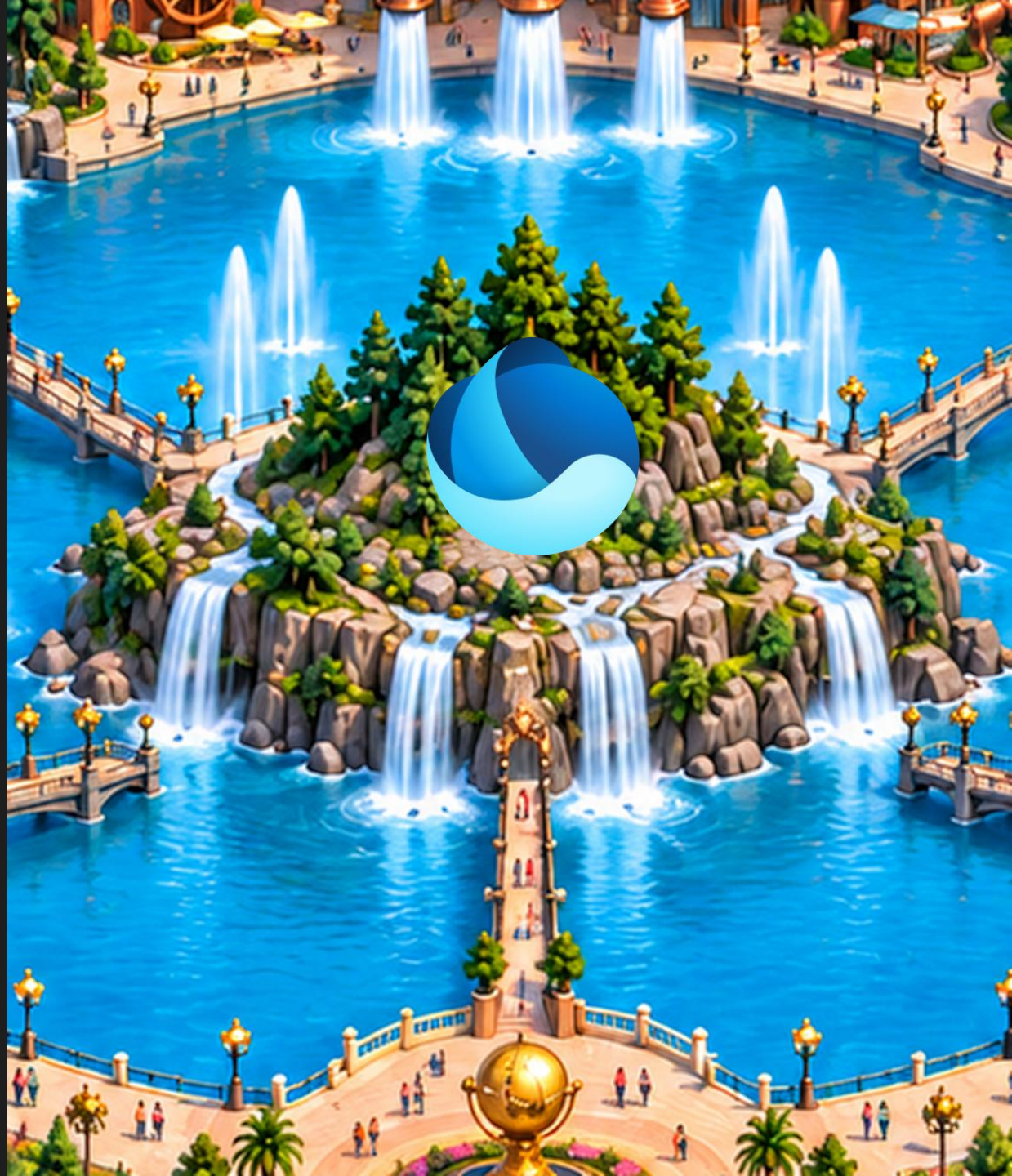


YOU ARE HERE

2 · THE LAKE

OneLake at the center

Shared analytical data. Several experiences can reuse it.



THE LAKE

Shared analytical data. Several experiences can reuse it.



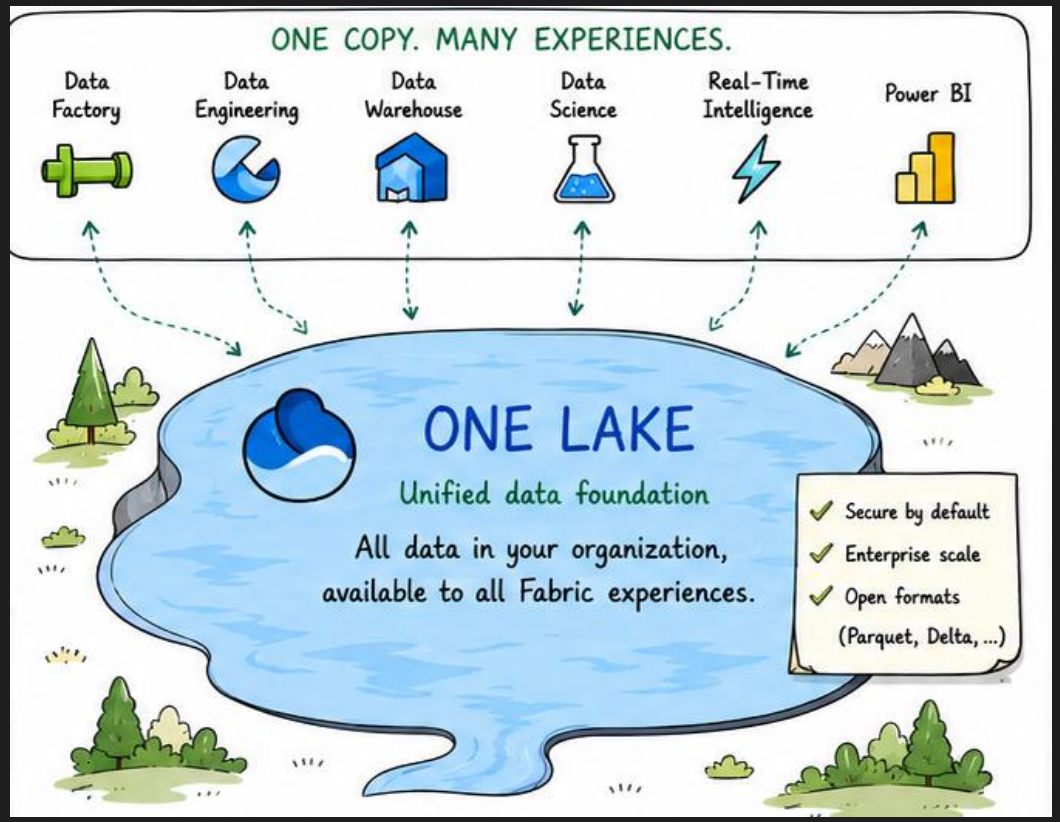
You don't have to take my word for it

The community drew the same map.

Fabric Notes is Microsoft's set of sketch explainers for the core concepts. Their OneLake note puts the same thing at the centre that we just did, with the workloads arranged around it — and gets there independently.

Shared data. Many experiences.

When two people draw a platform the same way, that is usually the shape of the platform.



Detail from "OneLake: One data lake for everything in Fabric" — Fabric Notes, Microsoft, CC BY 4.0 · aka.ms/fabricnotes

Files and tables in a lakehouse

A file upload and a queryable Delta table are different steps.

Files

A CSV or JSON drop lands here.

It remains a file until you load or transform it.

A notebook can read its contents.

Tables

dim_entity and fact_wait_snapshot live here as Delta.

Query them through the SQL analytics endpoint.

Use them in a semantic model.

Our collector notebook writes Delta tables directly. A CSV upload needs Load to tables first.

Follow one piece of data

SOURCE

Ride wait times

›

COLLECT

Notebook, hourly

›

STORE

Lakehouse

›

SHAPE

Warehouse views

›

SEE

Model and report

One ride's wait time, from the park's API to a Power BI report.

Shared storage and additional results

Several tools can use the same analytical tables.

Reuse

A notebook, SQL endpoint and model can work from shared analytical tables.

Shortcuts can reference data already stored elsewhere.

Additional results

A cleaned table is a new analytical output.

An Import model stores a cache.

Teams still define meaning and permissions.

Shared storage reduces copies. Data ownership and access rules still matter.

OneLake shortcuts

A shortcut references data in another storage location.

Why use one?

Reuse an existing table without maintaining another ingestion copy.

Example: use another team's shared table from your lakehouse.

What to check

Permissions and connection credentials.

A supported format for the tool you'll use.

Source availability and performance.

A shortcut points to existing data. It does not create an independent backup.

YOU ARE HERE

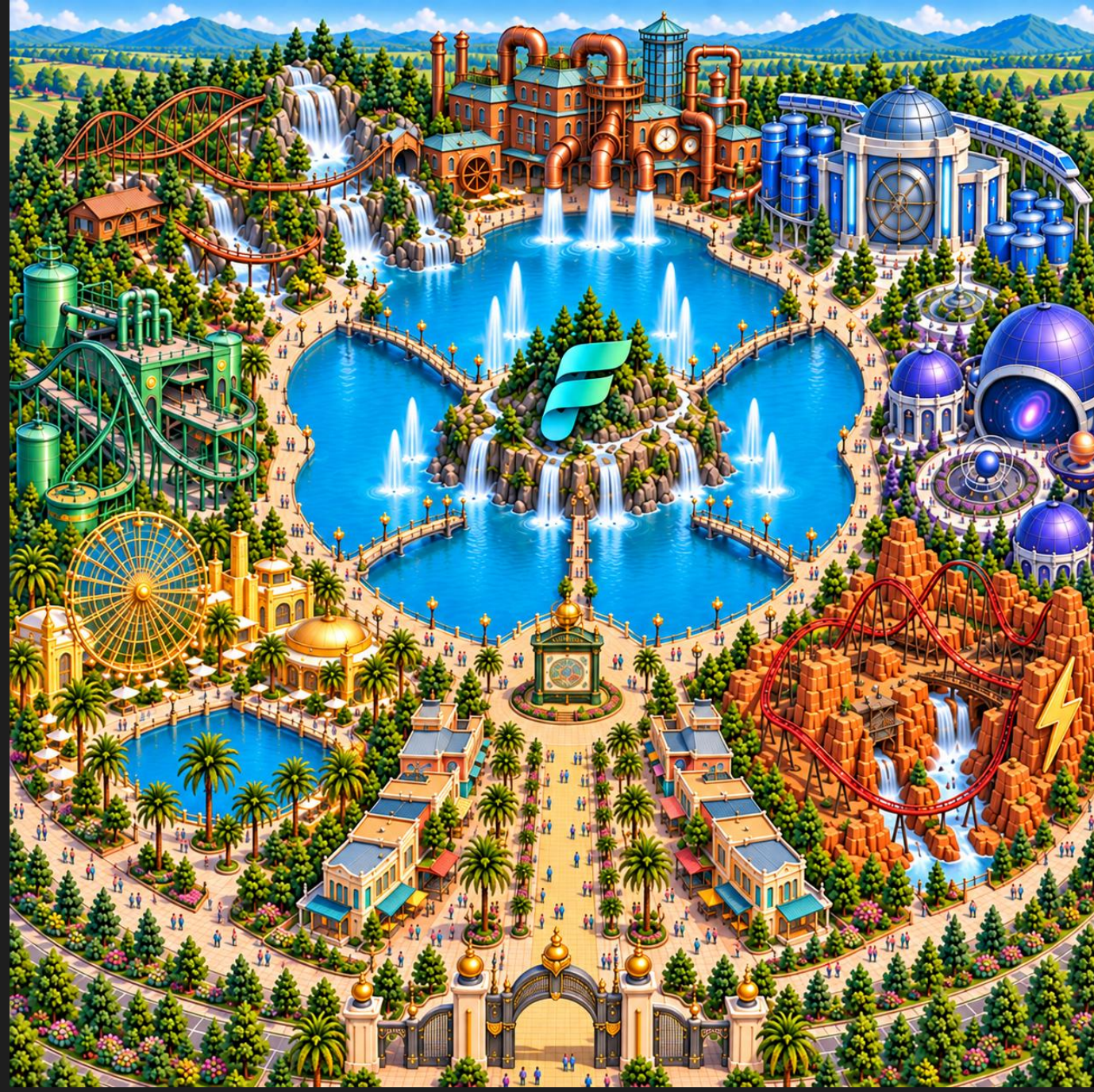
3 · THE LANDS

All seven, one walk

What exists, and where it sits. No decisions yet — those are section 4.

- **Power BI**
- **Data Factory**
- **Data Engineering**
- **Databases**
- **Data Warehouse**
- **Data Science**
- **Real-Time Intelligence**

These are the seven workloads we will meet today. Fabric continues to grow.



Power BI

The land that was here first. The park got built around it.

THE RIDES

- **Reports**
  The visuals people actually open
- **Semantic models**
  The measures and relationships underneath
- **Org apps**
  How you package reports for consumers

YOU ARE HERE



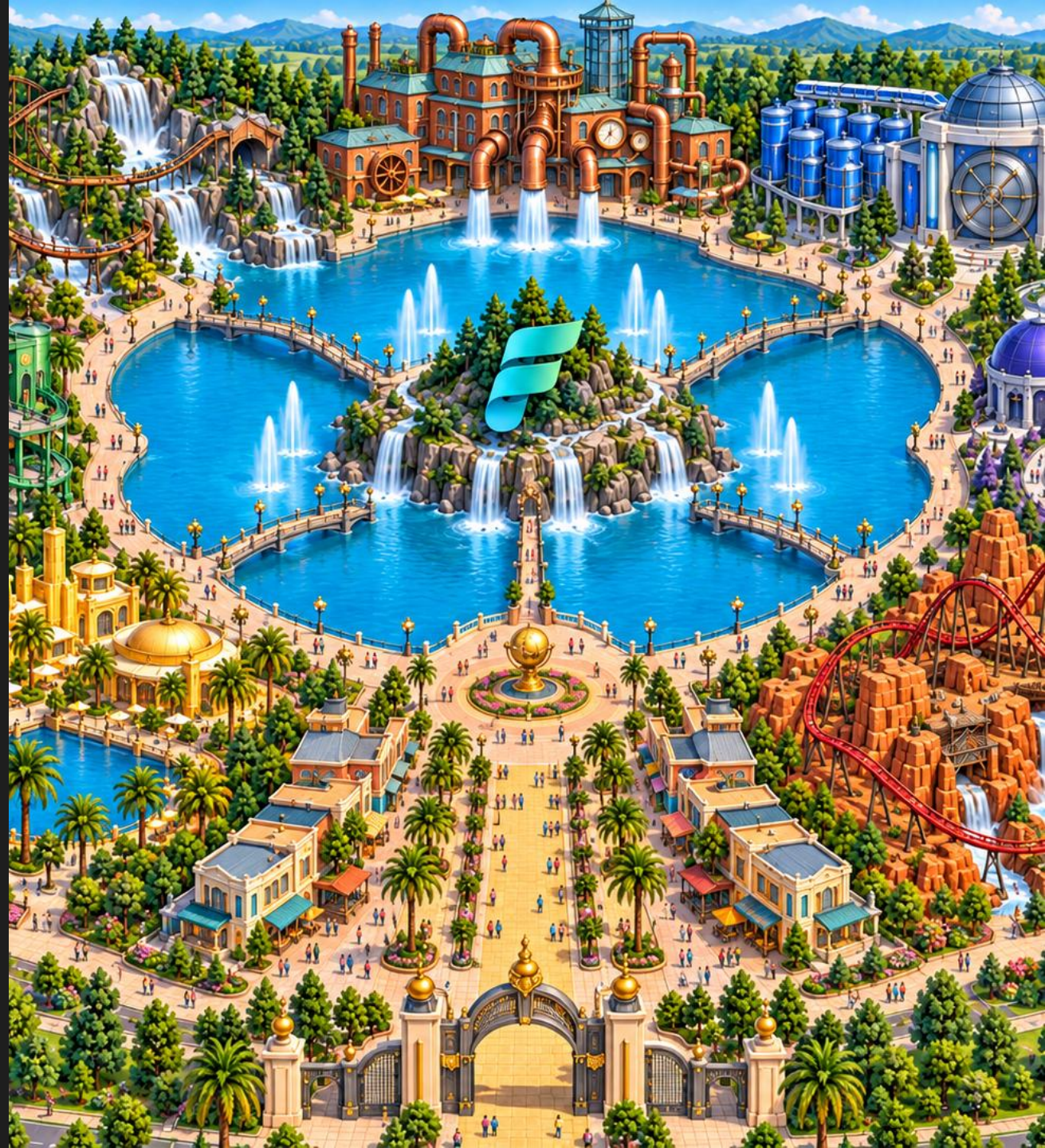
Direct Lake is a storage mode on a semantic model. It lives here, not in a land of its own.



LIVE DEMO 2

Ride Along

Reports



Data Factory

Getting data in, and deciding what runs when.

THE RIDES

- **Pipelines**
  Copy this, then run that, on a schedule
- **Dataflows Gen2**
  Power Query transforms, no code required
- **Copy job**
  Point-to-point movement without building a pipeline

YOU ARE HERE



If you have built a dataflow in Power BI, you have already ridden here.



Data Engineering

Spark and Python, working directly on the lake.

THE RIDES

- **Lakehouse**
  Files and Delta tables side by side
- **Notebooks**
  Where the code lives
- **Spark job definitions**
  Scheduled code without a notebook

YOU ARE HERE



The busiest ride in the park. Section 4 is where we decide whether it is yours.



Databases

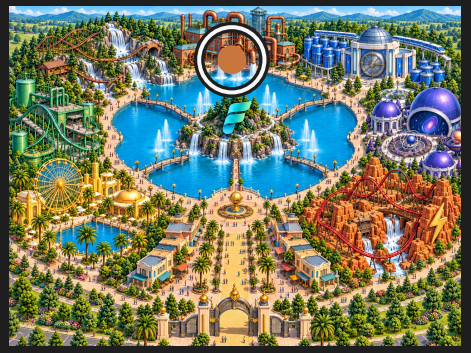
Operational data for applications and transactions.



THE RIDES

- **SQL database in Fabric**
  Relational data and application transactions
- **Cosmos DB in Fabric**
  NoSQL documents for flexible application data

YOU ARE HERE



Application data becomes available in OneLake for analytics.

Data Warehouse

Analytical SQL tables with supported T-SQL and transactions.

THE RIDES

- **Warehouse**
  Tables only, fully structured, writes included
- **SQL analytics endpoint**
  Read-only T-SQL over a Lakehouse
- **Mirrored databases**
  A live copy of an outside database, landed as Delta

YOU ARE HERE



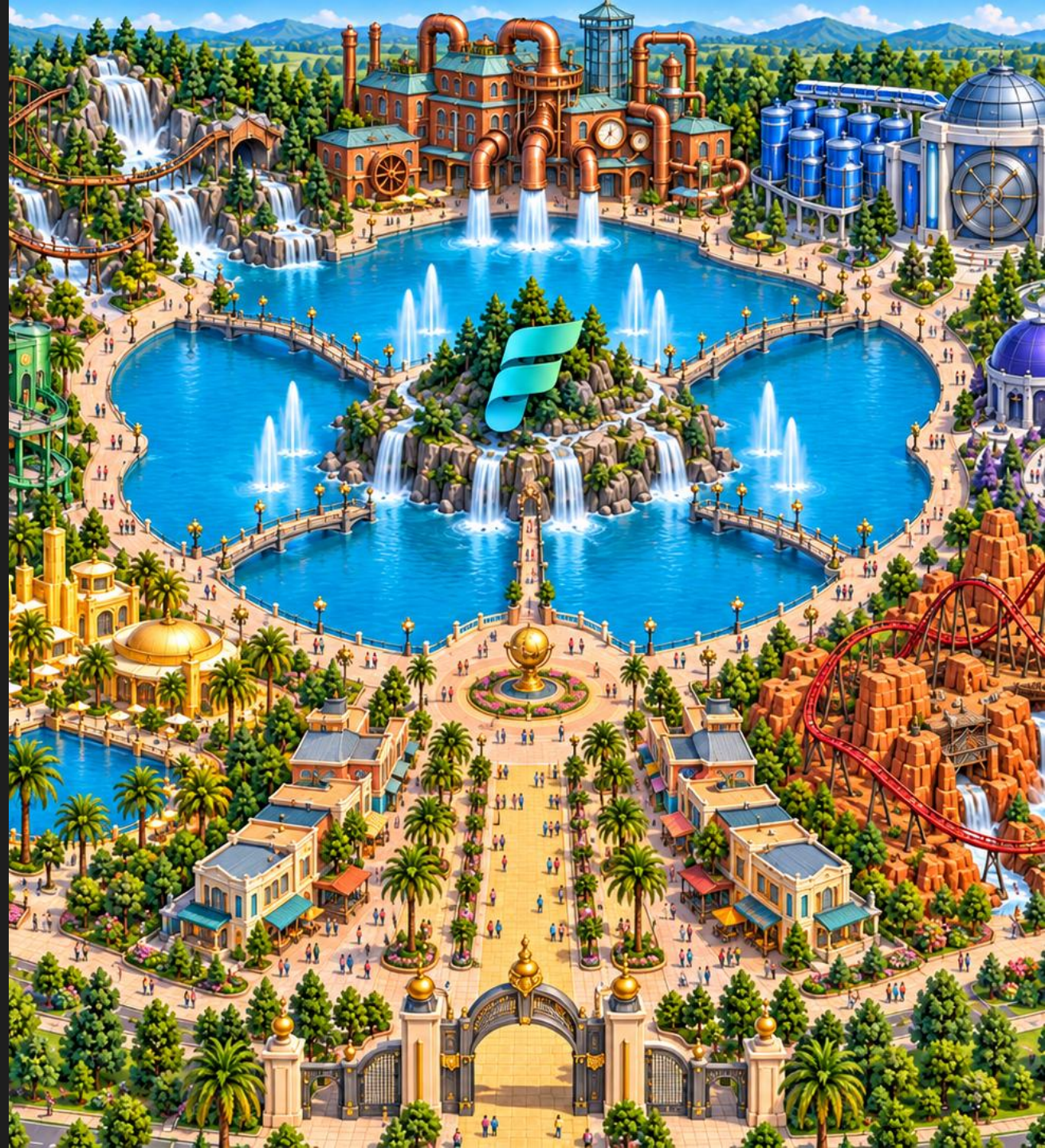
Existing SQL skills help. Check the supported T-SQL surface area.



LIVE DEMO 3

Ride Along

Behind the report



Data Science

Models and experiments, pointed at the data already in the lake.

THE RIDES

- **Notebooks**
  The same ride as Data Engineering, a different purpose
- **Experiments**
  Track runs and compare them
- **ML models**
  Register a model and score with it

YOU ARE HERE



Not on the critical path for a first Fabric project. Know it is here; come back later.



Real-Time Intelligence

For data that arrives continuously rather than in batches.

THE RIDES

- **Eventstream**
  Point a live feed at somewhere it can land — low-code
- **Eventhouse + KQL**
  Where the stream lands. Time-series data, queried with KQL
- **Activator**
  Watches the stream and fires when a condition is met

YOU ARE HERE



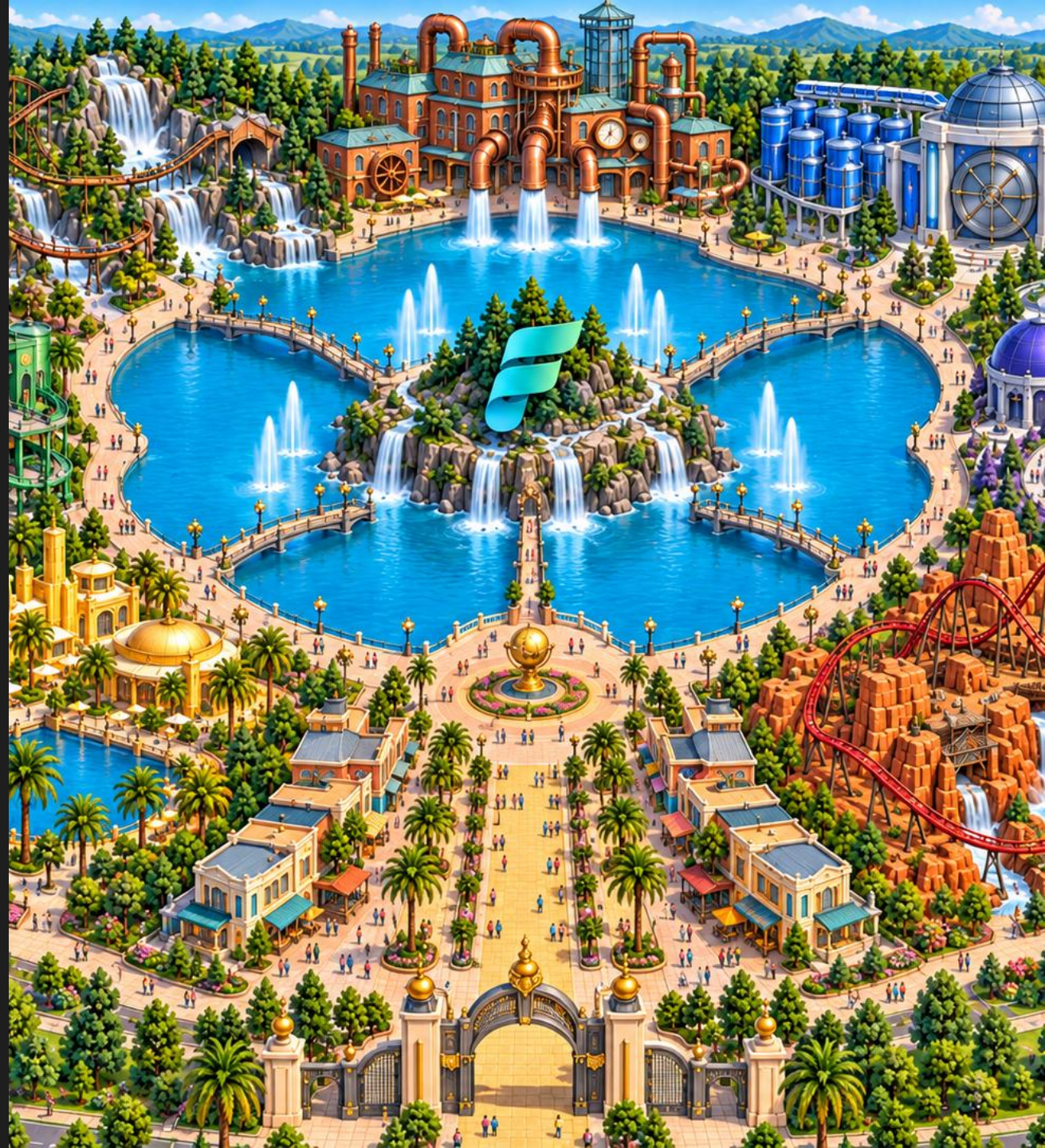
OneLake availability exposes a Delta representation; allow for processing latency.



LIVE DEMO 4

Ride Along

RTI



Keeping track of what you build

Version your work and promote changes between environments.

Git integration

GitHub or Azure DevOps

Version supported workspace item definitions.

Review changes and recover earlier versions.

Item definitions

Code, models and reports

Supported notebook code, semantic model and report definitions.

Lake data is stored separately.

Deployment pipelines

Development, test, production

Promote supported items between stage workspaces.

Check connections and settings for each environment.

Git records changes. Deployment pipelines move releases.

One word, three things

ORG APP

Packages reports for people who only read them.

WORKSPACE APP

The older way of doing the same job.

FABRIC APP

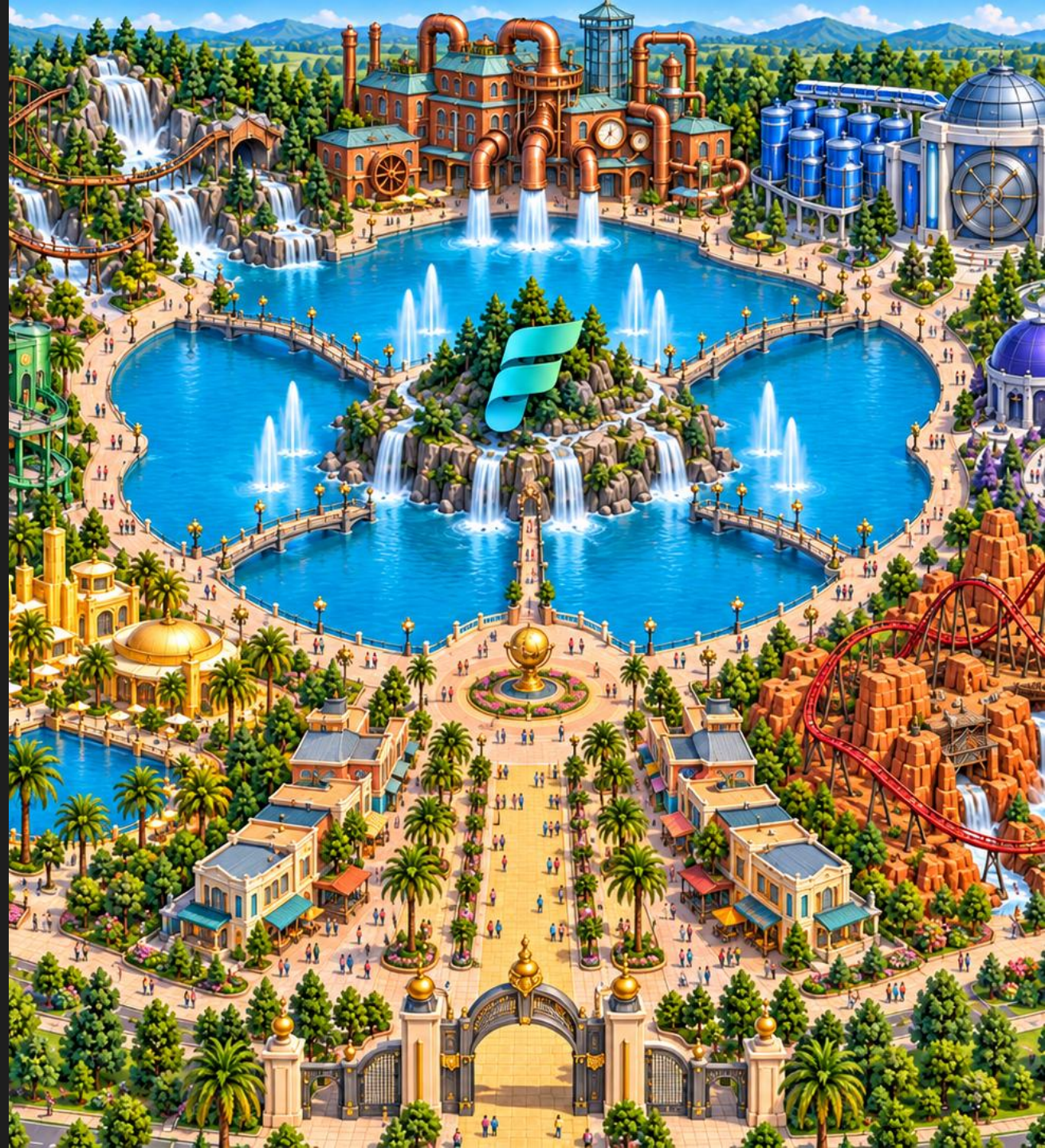
A real web application, built and hosted in Fabric. Preview.

Only the first one is a ride you take today.

LIVE DEMO 5

Ride Along

Fabric Apps



That was a tour, not a shopping list

Our wait-time route used a few rides, not all seven.

Data Engineering

Collect the waits

OneLake

Hold the data

Power BI

Model and report

Choose what solves your problem. The rest can wait.

YOU ARE HERE

4 · PICKING YOUR RIDE

Back at the map kiosk

What exists is settled. Now: which one do you actually walk to?



SQL database, Warehouse and Lakehouse

Operational transactions and analytical processing serve different needs.

ITEM	PRIMARY PURPOSE	WRITE PATH
SQL database	Application transactions	Operational SQL
Warehouse	SQL analytical tables	Supported T-SQL
Lakehouse	Files and Delta tables	Spark and ingestion tools

Warehouse supports T-SQL writes. The lakehouse SQL endpoint reads its tables.

Pipeline, Dataflow or Notebook

Choose the tool for the work you need to express and maintain.

TOOL	GOOD FIRST USE
Pipeline	Schedule work and manage dependencies or retries
Dataflow Gen2	Transform data with Power Query and write the results
Notebook	Explore data or write custom code — Python, Spark or T-SQL kernels

A notebook is not only Spark. The T-SQL kernel runs against a Warehouse or SQL analytics endpoint.
A pipeline can orchestrate a Dataflow or a Notebook.

A Notebook is a tool, not a rite of passage.

Power BI storage modes

The model's storage mode affects how it retrieves data.

MODE	HOW IT GETS DATA	FRESHNESS DEPENDENCY
Import	Queries data loaded into the model	Successful data refresh
DirectQuery	Sends queries to the source	Source and query behavior
Direct Lake	Loads columns from Delta tables	Successful model updates

Direct Lake still needs model updates, suitable capacity and a security design.

The map key

Five rides worth your first afternoon. Everything else on the map can wait.

1. **Lakehouse**
   in Data Engineering
2. **Warehouse**
   in Data Warehouse
3. **Notebook**
   in Data Engineering
4. **Pipeline**
   in Data Factory
5. **Semantic model**
   in Power BI



Changing tools and reusing data

Shared data can make a later change easier.

You may reuse

Accessible source files and analytical tables.

Business definitions and test cases.

Some queries and model concepts.

You may change

Transformation code and scheduling.

Table layout or loading method.

Connections and security rules.

A small first project gives you evidence for the next decision.

YOU ARE HERE

5 · THE TAKE-HOME MAP

Heading for the gate

What to do on Monday, and where to go next.



Pick one ride, not the whole park

If you live in Power BI

Start with a Lakehouse and a Direct Lake semantic model. It is the shortest distance from what you know to what is new.

If you live in T-SQL

Start with a Warehouse. Your SQL skills help; check the supported features and learn the lake as you go.

If you live in Python

Start with a Notebook against a Lakehouse. Same language you already write, new place to point it.

Nobody does the whole park in one day. Pick the gate nearest to where you already stand.

Your first report

Start with the demo data provided with this session.

STEP	ACTION
1 Workspace	Use a workspace with capacity and create a Lakehouse
2 Upload	Upload the supplied demo data to Files
3 Table	Load into Delta tables; check rows and column types
4 Model	Create a Direct Lake semantic model from the tables
5 Report	Build a visual that answers a question about the data

Check your result against the supplied demo data.

Your next visit: understand and build

Places that give a beginner a practical starting point

Workloads hub	Explore item types, samples and links inside Fabric.
Fabric Notes	Use the sketch explainers to understand the platform.
Paul Turley's Essentials	Follow the Fabric & Power BI Essentials series.
Fabric Forge	Learn through lessons, exercises and community calls.

Pick one next action for the ride you chose today.

Your next visit: practice and connect

Courses, working checklists and people to learn with

Tabular Editor Learn	Develop your Power BI semantic modeling skills.
Data Goblins checklists	Use a checklist while building your report or model.
Figuring Out Fabric	Hear practical conversations about learning Fabric.
Your local user group	Bring your first project and a question to a meetup.

Pick one next action for the ride you chose today.

Step inside THE CUBE



Community and courses · Free plan and paid memberships · guyinacube.com

THANK YOU

Jason Romans

Email jason@datashepherd.dev
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